AUTO WIPER SYSTEM

id091900000900

Outline

- The auto wiper system performs windshield wiper operation and stopping according to the rainfall amount on the windshield glass.
- The auto wiper system is equipped with intermittent, and continuous low and high operation.

Function

- The BCM controls the windshield wipers based on the windshield wiper operation signal from the rain sensor.
  - — Intermittent operation control
    - When the rain sensor detects an amount of rainfall, the windshield wipers are operated once at low speed.
    - With the rain sensor, the intermittent period of the windshield wiper operation is switched at 5 levels according to the amount of rainfall detected.
  - — Continuous low speed operation control
    - When the rain sensor detects an amount of rainfall greater than the amount required for intermittent operation, the windshield wipers are operated continuously at low speed.
  - — High speed operation control
    - If the rain sensor detects a vehicle speed of **4 km {2 mph} or more** and a rainfall amount sufficient for low continuous wiper operation control or more, the windshield wipers are operated **2 times** at high speed, and if a rainfall amount sufficient for low continuous operation control or more is detected, the windshield wipers are operated continuously at high speed.
    - If a vehicle speed of **4 km/h {2 mph} or less** and a rainfall amount sufficient for low continuous operation control or more is detected, the windshield wipers are operated continuously at low speed.
  - — Sensitivity adjustment function
    - The sensitivity adjustment function can adjust the rain sensor rain detection sensitivity by operation of the wiper and washer switch sensitivity adjustment volume at 5 steps.
    - When the windshield wipers are stopped during an interval and the sensitivity adjustment volume is operated in the direction of high sensitivity, the windshield wipers operate one time to notify the driver that the rain detection sensitivity has been changed.

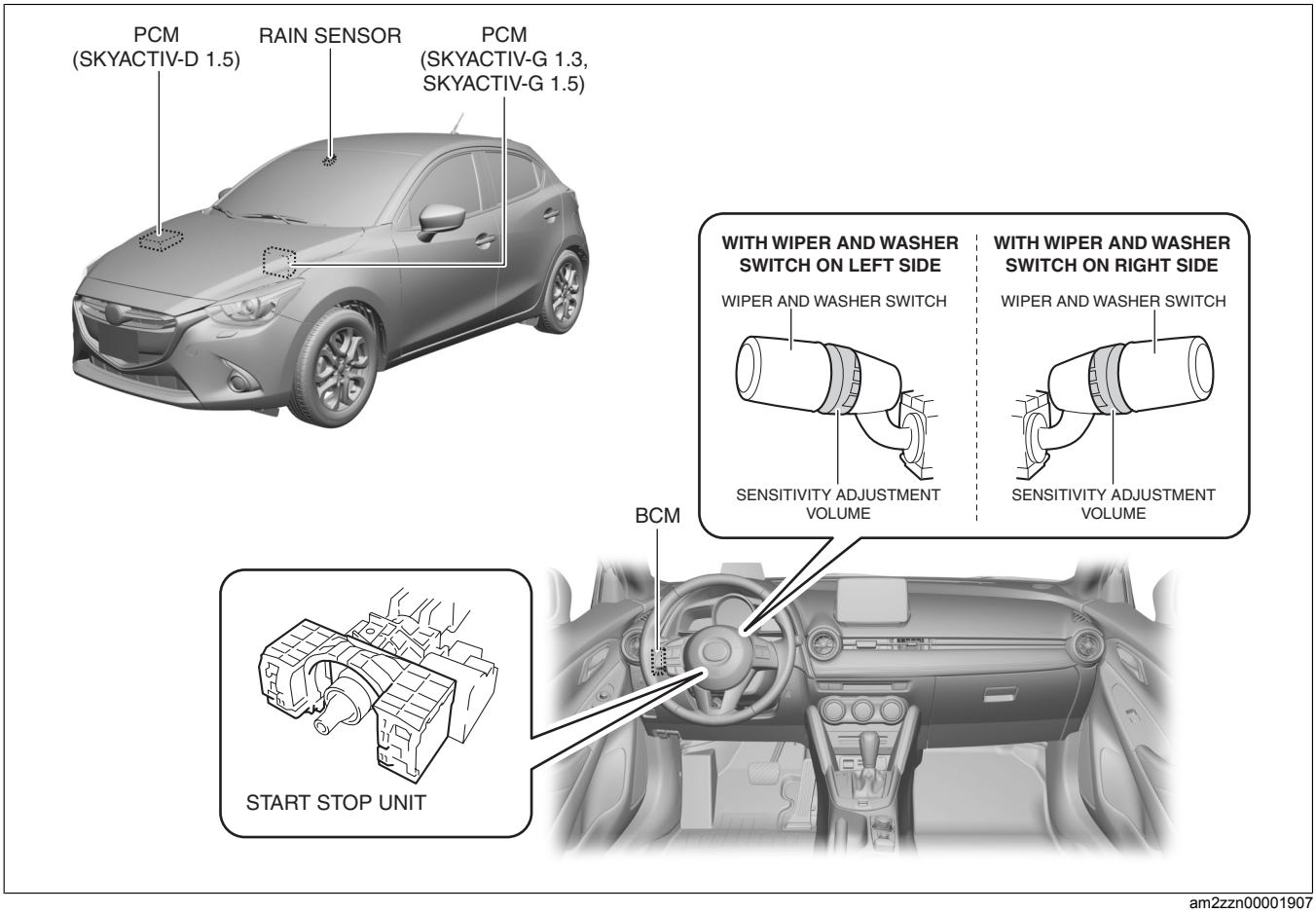
Personalization features

- A personalization feature has been adopted with which settings for the auto wiper system can be changed.
- When the auto wiper system is set to [Off (Intermittent)] or [Off] and the windshield wiper switch is in the AUTO position, the windshield wipers operate intermittently.
- For details on the personalization features, refer to the [WIPER/WASHER SYSTEM PERSONALIZATION]. (See WIPER/WASHER SYSTEM PERSONALIZATION.)

On-board diagnostic function

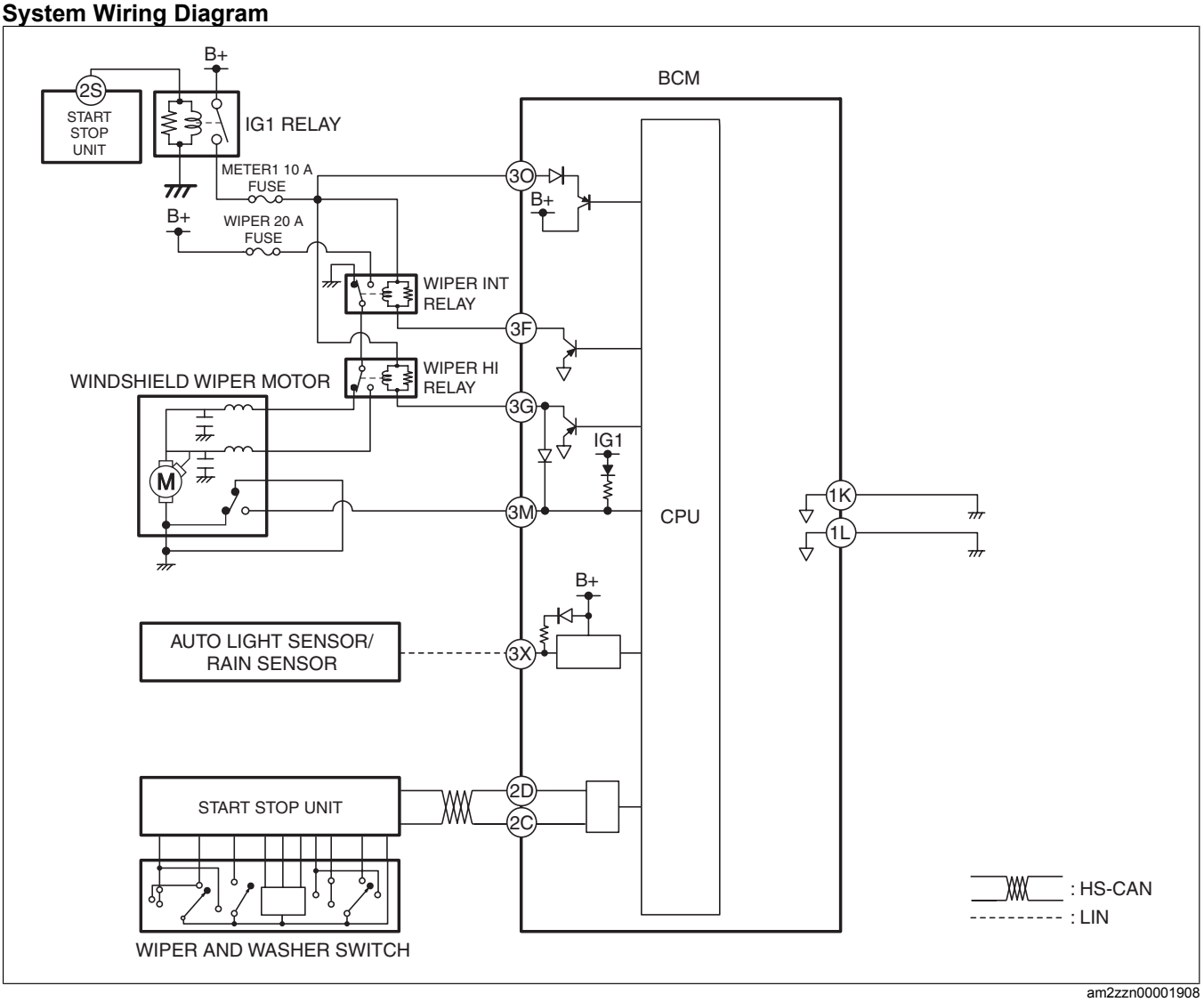
- If a malfunction occurs in the auto wiper control-related switch, sensor, and circuit, the BCM detects DTCs. For details, refer to the [ON-BOARD DIAGNOSTIC SYSTEM [BODY CONTROL MODULE (BCM)]] DTC table. (See ON-BOARD DIAGNOSTIC SYSTEM [BODY CONTROL MODULE (BCM)].)

Structural View



am2zzn00001907

System Wiring Diagram



am2zzn00001908

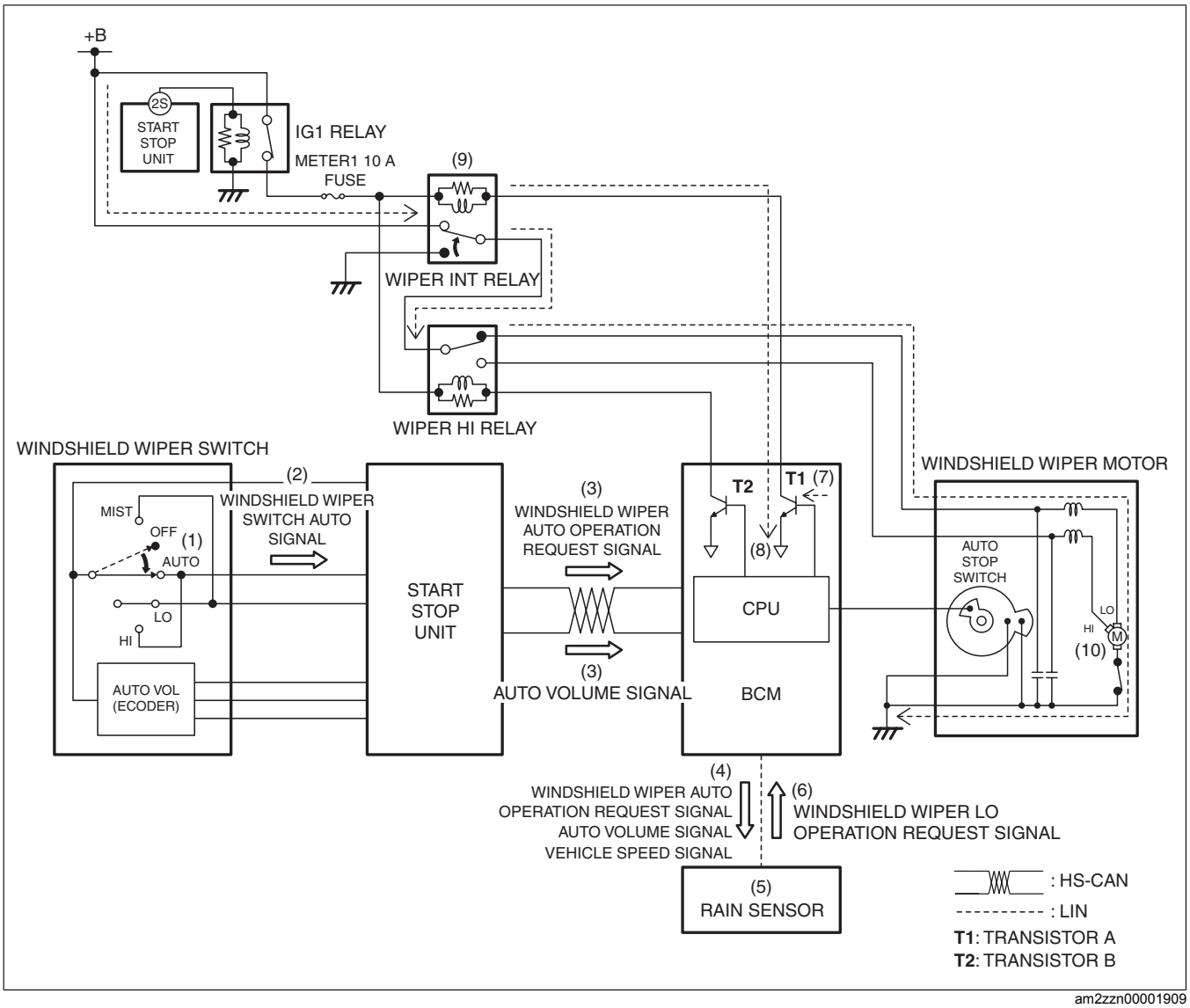
Operation

Warning

- **If servicing such as vehicle washing is performed with the ignition switched ON (engine off or on) and the windshield wiper switch is in the AUTO position, the windshield wipers may operate automatically and your finger or hand could be pinched, leading to an injury or wiper system malfunction. Always switch the ignition off or turn the windshield wiper switch to the OFF position before servicing.**

Intermittent operation/Continuous low speed operation

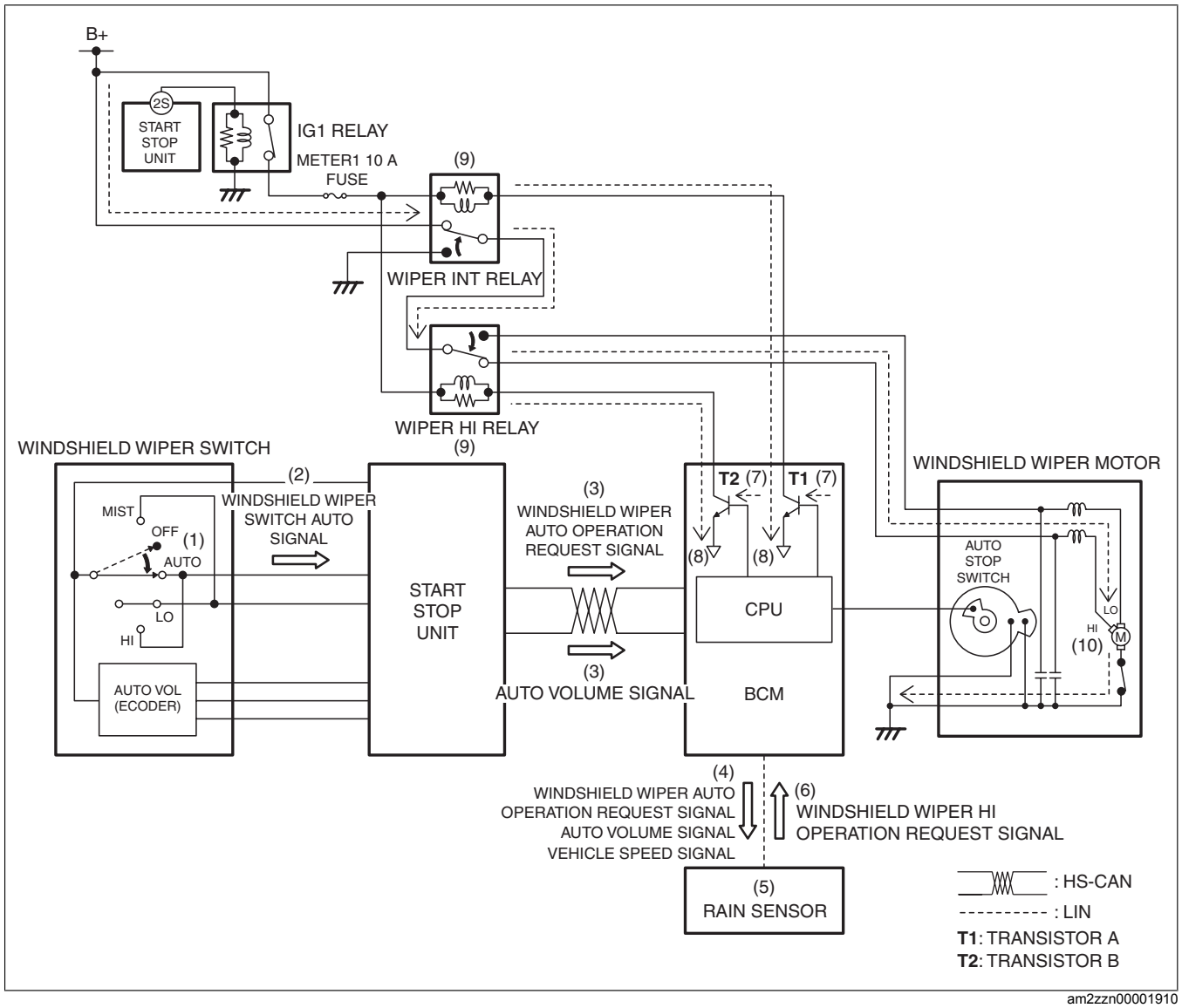
1. When the windshield wiper switch is in the AUTO position (1) with the ignition switched ON (engine off or on), the start stop unit detects (2) the windshield wiper switch AUTO signal.
2. When the start stop unit detects the windshield wiper switch AUTO signal, the windshield wiper AUTO operation request signal and the AUTO volume signal are sent (3) to the BCM by CAN communication.
3. The BCM sends (4) the windshield wiper AUTO operation request signal and AUTO volume signal, and the vehicle speed signal sent from the PCM to the rain sensor by LIN communication.
4. When the rain sensor receives the windshield wiper AUTO operation request signal, it detects (5) the amount of rainfall.
5. The rain sensor determines the operation interval of the windshield wipers based on the detected rainfall amount and illumination level, and sends (6) a windshield wiper LO operation request signal to the BCM.
6. When the BCM receives the windshield wiper LO operation request signal, it turns transistor A on (7).
7. When transistor A turns on, a ground circuit with the wiper INT relay is established (8), and the wiper INT relay turns on (9).
8. When the wiper INT relay turns on, the windshield wipers operate intermittently or at low continuous operation (10).



am2zzn00001909

High speed operation

1. When the windshield wiper switch is in the AUTO position (1) with the ignition switched ON (engine off or on), the start stop unit detects (2) windshield wiper switch AUTO signal.
2. When the start stop unit detects the windshield wiper switch AUTO signal, the windshield wiper AUTO operation request signal and the AUTO volume signal are sent (3) to the BCM by CAN communication.
3. The BCM sends (4) the windshield wiper AUTO operation request signal and AUTO volume signal, and the vehicle speed signal sent from the PCM to the rain sensor by LIN communication.
4. When the rain sensor receives the windshield wiper AUTO operation request signal, it detects (5) the amount of rainfall.
5. The rain sensor determines the operation interval of the windshield wipers based on the detected rainfall amount and illumination level, and sends (6) a windshield wiper HI operation request signal to the BCM.
6. When the BCM receives the windshield wiper HI operation request signal, it turns transistors A and B on (7).
7. When transistors A and B turn on, a ground circuit with the wiper INT relay and the wiper HI relay are established (8), and the wiper INT relay and wiper HI relay turn on (9).
8. When the wiper INT relay and wiper HI relay turn on, the windshield wipers operate at high speed (10).



am2zzn00001910